

Yuxiang Wei

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Education

PhD	Georgia Institute of Technology, <i>Electrical and Computer Engineering</i>	2028.5
	Research Focus: machine learning, computer vision, spatiotemporal models, brain imaging, MRI	
BS	University of Electronic Science and Technology of China, <i>Electronic Information Engineering</i>	2023.6
BS	University of Glasgow, <i>Electronics and Electrical Engineering</i>	2023.6

Skills

- **Programming and Tools:** Python, Pytorch, Sci-kit Learn, Numpy, C++, CUDA, Shell, Git, Linux, AWS.
- **Machine Learning:** SVM, linear regression, PCA, kernel methods, CNN, Transformer, SSM.

Experience

TReNDS Center, Georgia Institute of Technology - Graduate Research Assistant Atlanta, GA | 2023.8-present
Early Asymptomatic Alzheimer's Diagnosis with Knowledge Distillation

- Proposed an adaptive **knowledge distillation** framework to utilize the progression nature of Alzheimer's, which adaptively reuses information from the later stage to facilitate diagnosis of early asymptomatic Alzheimer's.
- Ideated and designed a margin-aware distillation loss that ensures better separability of the minority class, thereby improving the overall accuracy for over 10% with a highly imbalanced dataset.
- Collaborated with 2 lab members on the data collection of patients; responsible for cleaning and preprocessing. State-space Model for Universal Brain Network Analysis
- Built an end-to-end spatiotemporal **state-space model** to analyze dynamic brain networks based on functional MRI; designed a new aggregation method that maximally preserves the topology of brain networks.
- Conducted comprehensive comparative and ablation study on the model using multiple large-scale datasets; increased the prediction accuracy over different cognitive and behavioral characteristics for 5%.

Tencent - Applied Scientist Intern Shenzhen, China | 2024.5-2024.8

Multi-treatment Multi-task Uplift Modeling

- Explored **causal inference** and **uplift modeling** to boost user activity for one of the most popular mobile games from Tencent; proposed a novel two-model uplift modeling framework that explicitly models interactions between treatments and user features, and directly predict the uplift for multiple treatments and tasks.
- Discovered the bias of existing methods that results from non-random data collection; designed a new uplift estimation method to reduce bias. The model increases number of monthly users by over 10% under A/B test.
- Led the data collection team to discover real users' feedback on the new model and adjusted the treatment deployment strategy accordingly based on the proposed model.

HPC-AI Tech - Software Engineer Intern Beijing, China | 2022.4-2022.6

- Contributed to an open source distributed deep learning system named **ColossalAI**; developed an elastic DL system using **Ray**, which can flexibly schedule resources for data and pipeline parallelism and adjust the scale of parallelism automatically; increase the system's throughput by over 20%.
- Engaged in cloud deployment of the system with **Docker** and **Kubernetes** for better usability.

Golby Lab, Harvard Medical School - Undergraduate Research Assistant Boston, MA | 2022.6-2023.4

Multi-view Learning for Tabular diffusion MRI analysis

- Explored brain-based phenotypes prediction using white matter and gray matter MRI; developed **multi-view learning** framework and reduced the error rate by over 10%.

Tsinghua University - Undergraduate Research Assistant Beijing, China | 2020-2022

Multi-label Wafer Defect Recognition with Deep Learning

- Investigated and developed new deep learning models, including **transformer**, **convolution**, and **autoencoder** to enhance multi-label wafer defect recognition; improve the accuracy from 92% to 98%.
- Exploited **wavelet transform**'s potential in neural networks and developed its gradient propagation process.

Engineering Project

Automated Lecture Recording Summarization and Q&A System *personal project*

- Established a system using **LangChain** and **Retrieval-Augmented Generation**, which automatically transcripts lecture recordings and provides accurate responses to queries based on the course contents; utilized **Pinecone** as the primary vector dataset.

Publications

ORCID: 0000-0001-6552-8912

Google Scholar: scholar.google.com/citations?user=iGde1EAAAAAJ

- [1] **Yuxiang Wei**, Zhaoxin Qiu, Yingjie Li, Yuke Sun, Xiaoling Li, “Multi-Treatment Multi-Task Uplift Modeling for Enhancing User Growth.” Under Review by AAAI 2025.
- [2] **Yuxiang Wei**, Annes Abrol, Reihaneh Hassanzadeh, Vince D. Calhoun, “Hierarchical Spatio-Temporal State-Space Modeling for fMRI Analysis.” Under Review by AAAI 2025.
- [3] **Yuxiang Wei**, and Huan Wang. “Wavelet Integrated Attention Network with Multi-Resolution Frequency Learning for Mixed-Type Wafer Defect Recognition.” *Engineering Applications of Artificial Intelligence*, vol. 121, May 2023.
- [4] **Yuxiang Wei**, Yuqian Chen, Tengfei Xue, Fan Zhang, and Lauren J. O’Donnell. “An Explainable Multi-View Deep Network for Prediction of Non-Imaging Phenotypes using Anatomical Multi-view Data.” *The Medical Image Computing and Computer Assisted Intervention Society (MICCAI) workshop* 2023.
- [5] **Yuxiang Wei**, Tengfei Xue, Yogesh Rath, Nikos Makris, Fan Zhang, and Lauren J. O’Donnell. “Age Prediction Performance Varies Across Deep, Superficial, and Cerebellar White Matter Connections.” *IEEE Engineering in Medicine and Biology Society (EMBC)* 2024.
- [6] **Yuxiang Wei**, Anees Abrol, Vince D. Calhoun, “A deep spatio-temporal attention model of dynamic functional network connectivity shows sensitivity to Alzheimer’s in asymptomatic individuals.” *IEEE Engineering in Medicine and Biology Society (EMBC)* 2024.
- [7] **Yuxiang Wei**, Siqi Wang, Yun Li, “Graph Theory Based Machine Learning for Analog Circuit Design.” *International Conference on Automation and Computing (ICAC)* 2023.
- [8] **Yuxiang Wei**, and Huan Wang. “Mixed-Type Wafer Defect Recognition with Multi-Scale Information Fusion Transformer.” *IEEE Transactions on Semiconductor Manufacturing*, vol. 35, no. 2, pp. 341-352, May 2022.
- [9] **Yuxiang Wei**, and Huan Wang. “Mixed-type Wafer Defect Pattern Recognition Framework based on Multi-faceted Dynamic Convolution.” *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1-11, 2022.
- [10] Yitian Wang, **Yuxiang Wei**, and Huan Wang. “A Class Imbalanced Wafer Defect Classification Framework Based on Variational Autoencoder Generative Adversarial Network.” *Measurement Science and Technology*, 2022.